



Inflation risk and stock returns: Evidence from US aggregate and sectoral markets

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ABSTRACT

This study examines the relation between inflation risk and stock returns and finds a negative relation in US aggregate data. A comparable result is found in other G7 economies. Introducing the inflation-induced equity market volatility as an incremental variable consistently produces a negative effect on stock returns. The same finding holds true for monetary policy uncertainty, which also produces a negative relation. Ignoring these two elements will produce a biased estimator of inflation's effect on stock returns. Testing of sectoral stocks in the US market indicates that most sectors find a significant number of sectors exhibit negative inflation coefficients, confirming the Fama proxy hypothesis. The exception is the energy sector that presents a positive sign, indicating a hedging capacity against inflation. Including the interacting term between a change in monetary policy uncertainty and lagged equity market volatility, the evidence suggests an enhanced negative effect on stock returns.

1. Introduction

The issue of inflation and its impact on economic activity and stock market has recaptured the attention of households, investors, and policymakers after four decades of relatively stable variations in inflation. But many can remember when the annual inflation rate in the US climbed to 13.55% in 1980 (Amadeo, 2022), an increase that was mainly triggered by shocks from the oil crisis. Using a contractionary monetary policy, the Fed, however, has been able to achieve its target inflation rate of around the 2% level for a rather lengthy period of time. It was not until recently with the onslaught of the COVID-19 pandemic, which triggered high unemployment and considerable economic and social turmoil and subsequently precipitated a significant quantitative easing that the inflation rate surged, reaching 9.1% in June 2022. A further rise in inflation resulted from a sharp increase in energy prices due to the Russian-Ukraine conflict and, correspondingly, NATO's sanctions against the Russian economy. In an effort to bring down rampant inflation, the Fed started to hike interest rates in mid-March 2022, which created policy uncertainty and caused a significant selloff in stock markets as investors began to anticipate a higher discount rate and foresee a slowdown in future economic activity.¹ Viewed from this

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¹ While the author was drafting the original version of this paper, the Fed raised interest rates three times: +25 basis points (March 15–16, 2022), +50 basis points (May 3–4, 2022), and +75 basis points (June 14–15, 2022). It was forecasted that the Fed will hike the federal funds rate to 3.25–3.5 percent by the end of 2022 (Foster and Beers, 2022). A further rate hike is possible in the future Fed's meeting, depending on the data. While at the time of this paper's revision, the Fed had already raised the fed funds rate to a range of 5.25%–5.50% on the meeting of July 26, meeting, marking the 11th increase in the borrowing costs.

perspective, stock prices are observed to be negatively correlated with inflation despite the interventions of the Fed's monetary policy changes and uncertain geopolitical risk arising from the war threat.

A considerable amount of evidence, as notably reported by Bodie (1976), Nelson (1976), Fama (1981) and Schwert (1981), documents the negative relation between stock returns and inflation. These studies focus on a short-run approach known as Fama (1981) proxy hypothesis. On the other hand, there is also ample evidence of a positive relation between stock returns and inflation as documented by Boudoukh and Richardson (1993); Gultekin (1983), Solnik and Solnik (1997) and Toyoshima and Hamori (2011). These studies rely mainly on the Fisher (1930) hypothesis by emphasizing a long-run orientation. Some research even presents evidence of a mixed relationship as reported by Marshall (1992), Hess and Lee (1999), Kryzanowski and Rahman (2009) and Graham (1996). Different data sources, shock nature or monetary regimes most likely account for different directions for the sign of the coefficients in these studies.

Distinct reasons have also been offered for interpreting the test result variations. They include the use of different time periods for evaluating the model, different ways for generating the expected inflation, the use of different countries in conducting the test, and the application of different econometric models to test the hypothesis. Using a variation of the above approaches, this study develops an asset pricing-based model by incorporating inflation risk and inflation-induced equity market volatility, which reflects the current wisdom of modeling the time series approach (Chiang, 2023). This study is motivated by the following points.

First, high inflation often occurs in an extreme environment such as financial crisis, oil shocks, or periods of the threat of war among others. As a result, the time series will be interrupted and disturbed by some influential values, which ought to be controlled. Otherwise, the estimators will be subject to bias (Tsay, 1988; Pena, 2001). Most earlier studies in testing the return-inflation relation typically ignore this parametric impact.

Second, in observing the negative correlation between stock returns and inflation, most studies are based on Fama's (1981) notion that stock returns are positively related to real activity, while inflation is inversely related to real activity. However, Geske and Roll (1983) argue that changes in real activity help to reduce government debts. The monetization of the debt in turn reduces inflation. Their hypotheses successfully establish a negative linkage between stock returns and inflation through their connection to real activity. This study presents a new approach, however, in that the movements in the stock returns are often observed to be negatively correlated with the prevailing equity market volatility; stock market volatility exhibits comovement with inflation (Baker et al., 2022). Thus, putting these two statements together implies that the relation between stock returns and inflation is negatively correlated through their connections with equity market volatility (Chiang, 2023). This argument, however, has not been explicitly observed and fails to be incorporated in the discussion of conventional models used to test the relation of stock returns and inflation. When this view is combined with the traditional approach, the effects of inflation on stock returns can be seen not only by their direct impact on the erosion of income that reduces spending but also through the inflation-induced stock market volatility, which generates investor fears. The exclusion of the latter factor underestimates the damaging effect of inflation on stock returns.

Third, although substantial studies have been devoted to testing the relation between stock returns and inflation, the majority focus on the aggregate market and for the most part ignore an examination of individual sectoral stocks. Some sectors may potentially present a positive relation and, hence, have the capacity to hedge against inflation risk (Bouri et al., 2023).

Motivated by this gap in literature, this study examines stock returns and their relation to inflation by expanding the parametric effect as augmented by market volatility while controlling for conditional variance, money supply, and dividend yield and including dummy variables for influential observations. This paper contributes to the literature in the following aspects.

First, the model contains asset pricing features such that stock returns are positively correlated with conditional variance and support the risk-return trade-off hypothesis (French et al., 1987; Bali and Peng, 2006; Bali and Engle, 2010). Second, stock returns are negatively correlated with inflation in aggregate data (Fama, 1981) and most sectoral data. However, the energy sector, and sometimes the technology and basic materials stocks, demonstrate a positive sign and present an ability to hedge against inflation risk. Third, the data consistently show a negative relation between stock returns and inflation-induced equity market volatility. The significance of this variable indicates a potential for additional damage on stock returns arising from fears related to inflation-induced market volatility. The exclusion of this variable would produce biased estimators. Fourth, a rise (or decline) in the money supply or a decline (upward shift) in monetary policy uncertainty will have a positive (negative) effect on stock returns. Fifth, evidence shows that the interaction between the change in monetary policy uncertainty and lagged change in inflation equity market volatility is statistically significant. This variable provides a nonlinear term that enhances the policy effect on stock returns. Sixth, dividing inflation into upward and downward regimes produces evidence that shows a rise (decline) in inflation will produce a negative (positive) effect on stock returns. Seventh, the dummy variables for the periods of the 2008 global finance crisis (GFC) and 2021–2022 COVID-19 are negative and highly significant. The inclusion of these two dummy variables helps to alleviate the biasness of estimation. Eight, consistent with the valuation model, a rise in dividend yield enables us to predict a positive increase in stock returns.

The remainder of this paper is organized as follows. Section 2 provides a literature review. Section 3 presents an econometric model, which features a multiple regression model in a GED-GARCH process. Section 4 describes the data and related variables used in empirical estimations. Section 5 presents the empirical evidence for both aggregate and sectoral stock models. Section 6 conducts robustness tests for examining the impacts of nonlinear specifications. Section 7 concludes with the empirical findings and implication in practice.

2. Literature review

Evidence of a negative relation between stock returns and inflation is documented in the early literature (Bodie, 1976; Nelson, 1976, Fama and Schwert, 1977; Fama 1981; Schwert, 1981; Chen et al., 1986; Kaul, 1987; Amihud, 1996). One interpretation, known

as the money illusion effect as proposed by [Modigliani and Cohn \(1979\)](#), posits that a rational investor anticipates that cash flows move with inflation rate, rising in proportion to the discount rate; however, if cash flows are not adjusted by the same magnitude as the discount rate in inflation, stock prices will fall. [Fama \(1981\)](#) alternatively presents a “proxy effect” based on the following two relations. First, there is a positive association between stock returns and real activity; second, there is a negative correlation between inflation and real activity. Putting these two relations together implies that stock returns are negatively related to inflation ([Ram and Spencer, 1983](#)).² In fact, Fama’s notion of the proxy effect (1981) is based on the money demand theory and the quantity theory of money. To keep the money market in equilibrium, a rise in real activity and demand for money will be offset by a decline in inflation. [McCarthy et al. \(1990\)](#) test this hypothesis by examining data from the US, UK, and Germany and find evidence that stock returns are negatively correlated with inflation. conduct tests using a multivariate innovation decomposition and provide evidence that strongly supports the proxy hypothesis explanation in the US market. Recently, [Raghutla et al. \(2019\)](#) examined the Fama hypothesis using emerging market data and found a positive relation between stock returns and real output; their test also shows that output is significantly negative related to inflation. Thus, they document a negative relation between stock returns and inflation, supporting Fama’s hypothesis (1981).

[Geske and Roll \(1983\)](#) view the relation between stock returns and inflation in a different light. They argue that stock returns, which signal changes in real activity affect expected inflation, and further maintain rising output growth in the economy generates tax revenue and hence reduces government deficits that are monetized, lowering inflation. The study reported by [Mandelker and Tandon \(1985\)](#) finds a negative correlation between stock returns and expected inflation, which supports the [Geske and Roll \(1983\)](#) hypothesis. [James et al. \(1985\)](#) investigate the relations among stock returns, real activity, inflation, and money growth using a vector autoregressive moving average model (VARIMA) and find evidence that strongly confirms [Geske and Roll’s \(1983\)](#) reversed causality model. [Najand and Noronha \(1998\)](#) examine the direction of causal relations among stock returns, inflation, interest rates, and output in Japan. Their study indicates that inflation has a negative and significant effect on real stock returns at the lag of one month. However, inflation can also be explained by the lagged real stock return. Thus, there is a two-way causality between stock returns and inflation.

The Fisher hypothesis, which predicts a positive relation between nominal stock return and expected inflation, has received considerable attention. The viability of the Fisher hypothesis over a long run was tested by [Boudoukh & Richardson, 1993](#) using data for the US (1802–1990) and UK (1820–1988). Their investigation provides strong support for the presence of a positive relation between nominal stock returns and inflation over long horizons. In contrast, [Nelson \(1976\)](#) tests post-war data in the US, and cannot find evidence to support the Fisher hypothesis, but suggests that the presence of a negative relation between stock returns and both anticipated and unanticipated changes in the inflation rate. Similarly, [Gultekin \(1983\)](#) investigates the relation between common stock returns and inflation in 26 countries in the postwar period. The lack of support for the Fisher hypothesis in this investigation is reflected in the inconsistent positive relation between stock returns and inflation in the test results for most countries.

Yet, [Solnik and Solnik \(1997\)](#) using multi-country panel data find evidence that the Fisher model cannot be rejected at all horizons. Their study supports the result of [Boudoukh and Richardson \(1993\)](#) that the Fisher model holds at long horizons. Further studies by [Toyoshima and Hamori \(2011\)](#), who apply the data from the US, UK, and Japan, each arrive at the conclusion that confirms the Fisher hypothesis. [Hasan \(2008\)](#) examines the Fisher hypothesis using UK data and finds a positive and statistically significant relation between stock returns and inflation, suggesting that stocks can serve as a good hedge against inflation.

The literature also provides evidence of results with mixed signs subjected to regime conditions or sample periods. For instance, [Antonakakis et al. \(2017\)](#) reveal that correlations between the inflation and stock prices in the US evolve heterogeneously over time. [Marshall \(1992\)](#) reports that the negative relation between stock returns and expected inflation arises from real shocks, while the positive relation is attributable to monetary shocks. [Graham \(1996\)](#) examines the stock return and inflation relation based on subsamples with different monetary policy regimes. His study suggests that a negative relation is displayed when monetary policy is counter-cyclical or neutral and positive when the monetary policy is procyclical. In their study of the UK, Japan, and Germany, [Hess and Lee \(1999\)](#) find that “demand” shocks may lose their effect in the long run and “supply” shocks are likely to have long lasting effects. Further investigation by [Kryzanowski and Rahman \(2009\)](#) on the sources of shocks finds evidence that the effect of real shocks is more pronounced than that of monetary shocks. Using a Markov-switching approach to analyze data for Mexico and South Africa, [Cifter \(2015\)](#) shows that the relation between real stock return and inflation is negative during a recession. [Al-khazali \(2003\)](#) examines stock prices and inflation for 21 emerging capital markets and finds that in the short run a negative relation persists for all countries except Malaysia. His research, however, supports the Fisher effect in the long run. On the other hand, in a subsequent study that examines the Malaysian market in the post-1997 period, [Majid \(2010\)](#) finds evidence to support the Fisher hypothesis, contradicting the [Fama \(1981\)](#) proxy effect, which reveals a negative relation between real stock returns and inflation. This finding implies that the Malaysian stock market provides a good hedge against inflation. [Wong and Wu \(2003\)](#) using monthly data from G7 and eight Asian countries find evidence to support the Fisher hypothesis, showing a positive relation between stock returns and expected inflation for a long horizon but not between stock returns and contemporaneous inflation. [Li et al. \(2010\)](#), who examine the UK stock market, fail to find that stocks have the capacity to hedge against inflation in the short term. However, in the medium term, results are mixed, which is consistent with [Antonakakis et al. \(2017\)](#) in the US long historical study. Results from different inflationary periods show that the relation between inflation and returns varies in different regimes.³

² Fama’s (1981) statement has been questioned by those who ignore of the phenomenon of Phillips curve, which argues a positive relationship between inflation and real output.

³ The intention of this section is to provide a thorough rather than exhaustive review of the literature. [Madadpour and Asgari \(2019\)](#) provide a more comprehensive literature review of early studies.

The above literature review provides three important insights for directing new research. First, the valuation theory suggests that if inflation is higher, the cash flow and the discount factor are both adjusted upward accordingly, leaving the present value unchanged. Thus, the observed negative relation between inflation and stock returns should be attributable to investors' money illusion, unobserved market frictions, a lack of information to predict the inflation rate, uncertainty of monetary policy or simply a missing variable problem that leads to a misspecification of the test model (Azar, 2010).

Second, using the Fisher equation (1930) to frame the stock return and inflation relation, certainty is based on a long run approach, which may be subject to an equilibrium analysis and ignore the features of short-run dynamics. For investors concerned about hedging against inflation risk, it would be appropriate to use an asset pricing model that captures the inflation risk. Along this line, Sarte (1998) examines the Fisher equation by considering a consumption-based asset pricing model to investigate the existence of an inflation risk premium. The study shows that historical estimates of an inflation risk premium are trivial. Sarte (1998) argues that the insignificance of the inflation risk is due to the fact that surprises in inflation and unexpected movements in consumption growth exhibit little covariation in US data. In departing from consumption-based CAPM, this study uses an asset market-based model that emphasizes the information of inflation-induced equity market volatility, thereby eliciting inflation fears that could have a significant impact on stock returns. Hence, the study of the covariance between equity market volatility and inflation could shed additional light on the impact of inflation risk on stock returns.

Third, it matters little from an econometric point of view whether the test equation is based on a single equation reduced-form model as used by Fama (1981), Geske and Roll (1983), Solnick (1983) or a vector autoregressive model (VAR) of James, Koreisha, & Partch, 1985; Lee, 2010 and AlKhazali et al. (2004), since the elements included are mainly economic variables, such as stock returns, real activity, inflation and money supply. In previous investigations, tests of the relation between stock returns and inflation have not considered the risk factor, whether it is associated with conditional variance from stock market or from monetary policy uncertainty as an explanatory variable.

Moreover, from a time series analysis perspective, influential observations such as the time period involving the impact of the GFC or COVID-19 pandemic are not explicitly considered in the modeling process. This paper addresses the above issues as part of its examination of the relation between the stock returns and inflation relation (Cheema et al., 2020).

3. Econometric model

3.1. Basic model

The above discussions of an augmented market-based model can be concisely represented by a regression model expressed as:

$$R_t = C + \beta_1 \hat{\sigma}_t^2 + \beta_2 Inf_t + \beta_3 \Delta emv_{inf,t} + \beta_4 \Delta m_t + \beta_5 \Delta mpu_t + \beta_6 dy_{t-1} + \beta_7 D_{Crisis} + \beta_8 D_{COVID-19} + \varepsilon_t \quad (1)$$

where $\beta_1 > 0$, $\beta_2 < 0$, $\beta_3 < 0$, $\beta_4 > 0$, $\beta_5 < 0$, $\beta_6 > 0$, $\beta_7 < 0$, and $\beta_8 < 0$. The error term, $\varepsilon_t | I_{t-1} \sim GED(0, \sigma_{t-1}^2, \nu)$.

Equation (1) states that a stock return is a linear function of its own conditional variance from a GARCH-M process, $\hat{\sigma}_t^2$ (Merton, 1980; French et al., 1987; Bollerslev et al., 1992), inflation, Inf_t (Fama and Schwert, 1977; Fama, 1981; Schwert, 1981), a change in inflation equity market volatility, $\Delta emv_{inf,t}$ (Baker et al., 2022; Chiang, 2023), a change in money supply, Δm_t (Fama, 1981; Ram and Spencer, 1983; Sellin, 2001), a change in monetary policy uncertainty, Δmpu_t (Husted et al., 2020; Chiang, 2021a), and lagged dividend yield in natural logarithm, dy_{t-1} (Bali et al., 2009) while controlling economic shocks due to 2008 GFC and 2020 COVID-19 pandemic using dummy variables, D_{Crisis} (September 2008 – April 2009) and $D_{COVID-19}$ (March -April 2020) (Cheema et al., 2020), which, respectively, set values to be unity in the event dates present and zero otherwise. The choice of these explanatory variables is derived from recent empirical literature and reflects an attempt to provide new insights of the behavioral relationship and to reduce the biasness arising from ignoring influential observations (Tsay, 1988; Pena, 2001).

To complete the model, the variance equation is assumed to follow a GARCH(1,1) process (Bollerslev et al., 1992) expressed as:

$$\sigma_{i,t}^2 = \omega_0 + \omega_1 \varepsilon_{i,t-1}^2 + \omega_2 \sigma_{i,t-1}^2 \quad (2)$$

As noted by Nelson (1991), the error term is appropriate to follow the GED distribution, which is expressed as:

$$\varepsilon_{i,t} | \Phi_{t-1} \sim GED\left(0, \sigma_{i,t-1}^2, \nu\right) \quad (3)$$

The rationale for using this GED distribution stems from the error series, which can be smoothly transformed to deal with leptokurtotic distribution (fat tails) or a platykurtotic distribution (thin tails). It follows that the GED can work with the thickness of the tails of a distribution, especially in its ability to model the fat-tail series (Nelson, 1991; Li et al., 2005; Chiang, 2020).

⁴ The use of GARCH(1,1) was due to French et al. (1987) and popularized by Bollerslev et al. (1992) as a way to achieve a better fit of the stock return equation. Bollerslev (2010) summarizes different specifications of ARCH-type models. Note that Equation (2) does not include the asymmetric term for the lagged shock due to statistically insignificant.

Table 1

Summary statistics of US stock returns: Sample period is January 1990 – February 2022.

	R_TTMK	R_FINL	R_HLCA	R_INDU	R_REIT	R_CNSG	R_BMAT	R_ENGY	R_IVTS	R_TECH	INF
Mean	0.8665	0.8226	0.9548	0.9206	0.8852	0.8803	0.7597	0.6846	1.1442	1.1479	0.2100
Median	1.3469	1.5238	1.3716	1.4136	1.5542	1.1144	1.0257	0.9142	1.6592	1.7288	0.2102
Maximum	12.5248	16.9542	15.2145	15.9309	27.0084	14.8764	20.3893	28.5131	28.7664	19.5327	1.36756
Minimum	−18.7986	−24.0551	−13.0219	−22.6083	−35.8510	−13.1303	−28.8247	−45.5019	−39.4146	−32.3327	−1.7864
Std. Dev.	4.2233	5.6183	4.1045	5.1585	5.6728	3.8544	6.0674	6.2676	8.3962	7.0095	0.2687
Skewness	−0.7619	−0.9727	−0.3879	−0.6927	−1.3286	−0.3955	−0.6331	−0.8965	−0.5204	−0.6812	−1.1675
Kurtosis	4.8474	6.6265	3.7860	5.2375	10.7277	4.5221	5.7670	11.3476	4.6673	4.8988	12.867
Jarque-Bera	92.24	272.38	19.62	111.39	1074.01	47.57	148.92	1172.42	62.13	87.84	1653.74
Observations	386	386	386	386	386	388	386	386	386	386	386

Notes: Jarque-Bera statistics (JB) is a joint test for normality. $JB = n \left(\frac{(Sk)^2}{6} + \frac{(Ku)^2}{24} \right) \sim \chi^2(2)$, where Sk stands for skewness and Ku for kurtosis. The JB statistic is distributed as a chi-squared random variable with two degrees of freedom in large samples. If $JB < \chi^2_2$ for the critical value at the 0.05 level, normality is not rejected. Since the critical point of $\chi^2(2)$ at 5% level is 5.99, statistics strongly suggest the rejection of normality in all of markets, indicating the use of GED-GARCH procedure is well justified. The TTMK refers to the aggregate stock market. The sectoral returns are Financials (FINL), Health cares (HLCA), Industrials (INDU), Real estate (REIT), Consumption goods (CNSG), Retails (RTAL), Basic Materials (BMAT), Energy (ENGY), Investment services (INVS), and Technology (TECH). INF denotes inflation.

3.2. Testable hypotheses

Equation (1) contains information that is pertinent to an explanation of stock returns. Since the effect from $\hat{\sigma}_t^2$, Δm_t , $\Delta mp_{u,t}$, and dy_{t-1} have been widely discussed in the literature, here we only focus the effects of inflation on stock returns.

(i) *Short-run effect of inflation and stock returns.* This hypothesis can be simply set up based on a positive relation between stock returns and output growth, and a negative relation between output growth inflation as stated by Fama (1981). The inverse relation between output growth and inflation is essentially based on the money demand function. This is because an increase in demand for money balance due to a rise in output can only be offset by lower inflation in order to maintain money market equilibrium if the monetary authority does not accommodate an increase in money supply. Alternatively, Geske and Roll (1983) argue that an increase in output helps to raise tax revenue and reduce debts through monetization, resulting in lower inflation. Thus, stock returns are negatively related to inflation.

Viewed from a firm's perspective, higher inflation is considered to have a negative effect on stock returns, since it reduces expectations of corporate profits as higher inflation leads to higher input costs for business firms. In the meantime, a rise in inflation would reduce purchasing power of real cash balances, which causes households to cut consumption, resulting in a decline in sales and profits of businesses. Anticipation of lower cash flows due to inflation together with a rise in discount factor inevitably causes stock prices to fall.

(ii) *Inflation risk induced equity market volatility.* This hypothesis also postulates a negative relation between inflation-induced volatility and stock returns. The sensitivity of stock prices to news is well recognized and causes stock returns to be negatively correlated to equity market volatility; however, the equity market volatility is positively correlated with the inflation risk (Baker et al., 2022) since heightened inflation will prompt fears that the Fed will raise interest rates in order to curb inflation and maintain a stable currency value. Thus, an inflation-fear-volatility effect serves as an additional channel to illustrate the negative relation between inflation and stock returns that differs from the negative effect arising from the inflation erosion of the real income.

4. Data

The data in this study cover aggregate data for the US (and other G7 countries in Appendix) and ten sectoral stock indices and dividend yields, including sectors of Financials (FINL), Health cares (HLCA), Industrials (INDU), Real estate (REIT), Consumption goods (CNSG), Retails (RTAL), Basic Materials (BMAT), Energy (ENGY), Investment services (INVS), and Technology (TECH). The stock indices are the total return index (*RI* in *Datastream*), which includes dividends, interest, and rights offerings over a given month. The stock returns are measured by the natural log-difference of the *RI* index times 100. The monthly stock returns are reported in Table 1, which shows the average returns range from 0.6846 (ENGY) to 1.1442 (TECH); a similar performance is shown in the median statistics, which range from 0.9142 (ENGY) to 1.7288 (TECH). The standard deviations indicate that the CNSG sector (3.8544) has the lowest risk compared with the highest risk sector INVS (8.3962). The Jarque-Bera statistics (JB) for examining a joint test for normality are significantly large when compared against the critical point at the 5% level (=5.99 with 2 degrees of freedom), indicating that the normality of stock return series distribution should be rejected, and suggesting the use of GED-GARCH procedure is well justified.

All data for stock price indices, money supply (M2), CPI and dividend yields are monthly observations spanning from January 1990 through February 2022. Stock indices and dividend yields are downloaded from the database of *DataStream*; money supply and CPI are downloaded from economic database of Federal Reserve Bank of St. Louis.⁵

The monetary policy uncertainty (MPU) index is derived from the results of Access World News database of over 2,000 US newspapers (Baker et al., 2016, 2022). The monetary policy uncertainty is based on words related to monetary policy: {monetary policy, money supply, open market operations, fed funds rate, discount window, quantitative easing, forward guidance, interest on reserves, taper tantrum, Fed chair, Greenspan, Bernanke, Volker, Yellen, Draghi, Kuroda, Jerome Powell, lender of last resort, central bank, federal reserve, the fed, European Central Bank, ECB, Bank of England, bank of Japan, people's bank of China, pboc, pbc, central bank of China, Bank of Italy, Bundesbank}.

Baker et al. (2022) also construct a newspaper-based Equity Market Volatility (EMV) tracker that moves with the CBOE Volatility Index (VIX) and with realized S&P 500 volatility of returns. To construct the Inflation EMV tracker, $EMV_{inf,t}$, Baker et al. (2022) specify terms in following sets:

- E: {economic, economy, financial}
- M: {"stock market", equity, equities, "Standard and Poors" (and variants)}
- V: {volatility, volatile, uncertain, uncertainty, risk, risky}
- Inflation: {cpi, inflation, consumer prices, ppi, producer prices}

The formula used to measure the importance of inflation in equity market volatility during month t is given by $EMV_{inf,t}$, which is defined as:

⁵ We conduct unit root test using Augmented Dickey-Fuller test on the first differenced "ln(CPI)" and find an estimated statistic of -11.5305, which is significant at the 1% lever (-3.4471). This result suggests that the inflation series is stationary, that is, the inflation series is I(0). Note that the ln(CPI) will be I(1) to achieve stationary.

Table 2
Correlation between inflation and stock returns.

Correlation/t	INF	R_FINL	R_HLCA	R_INDU	R_REIT	R_CNSG	R_BASM	R_ENGY	R_IVTS	R_TECH
INF	1									
R_FINL	0.0142 0.28	1								
R_HLCA	−0.0217 −0.42	0.6632 17.36	1							
R_INDU	−0.0037 −0.07	0.8094 27.01	0.6359 16.15	1						
R_REIT	0.0174 0.34	0.6884 18.60	0.4887 10.98	0.6038 14.84	1					
R_CSTA	0.0010 0.02	0.6470 16.63	0.8049 26.58	0.5875 14.23	0.5078 11.55	1				
R_BASM	0.0480 0.94	0.7106 19.79	0.5481 12.84	0.8060 26.68	0.5711 13.63	0.5383 12.52	1			
R_ENGY	0.1044 2.06	0.5562 13.12	0.3882 8.25	0.6019 14.77	0.3916 8.34	0.3922 8.35	0.6575 17.10	1		
R_IVTS	0.0412 0.81	0.7971 25.87	0.5357 12.43	0.7342 21.19	0.4709 10.46	0.4631 16.09	0.6345 9.91	0.4512 9.91	1	
R_TECH	0.0282 0.55	0.5589 13.21	0.4654 10.31	0.7335 21.15	0.3784 8.01	0.3533 7.40	0.5633 13.36	0.3783 8.01	0.6919 18.78	1

Notes: This table presents correlations between the inflation and sectoral stock returns. The first row is the estimated coefficients, and the second row is t-statistics. INF denotes inflation; R refers stock return for sectors of Financials (FINL), Health cares (HLCA), Industrials (INDU), Real estate (REIT), Consumption goods (CNSG), Retails (RTAL), Basic Materials (BMAT), Energy (ENGY), Investment services (INVS), and Technology (TECH).

$$EMV_{inf,t} = \left(\frac{\#\{E \cap M \cap V \cap Inflation\}_t}{\#\{E \cap M \cap V\}_t} \right) EMV_t \quad (4)$$

where # denotes the count of newspaper articles in the indicated set, and EMV_t is the value of the overall EMV tracker in month t . For more information on Baker's methods, term lists for each category, and a discussion of how terms were selected, see "Policy News and Equity Market Volatility" (Baker et al., 2022).

5. Empirical evidence

The correlations of inflation with each sectoral returns are reported in Table 2. The US inflation rate as measured by the natural log-difference of the US consumer price index times 100. For each month it is 0.21% and is reported in Table 1. As shown in Table 2, eight out of 10 sectoral stock returns present positive correlations with inflation; only HLCA and INDU sectors exhibit negative signs. Media news often reports these correlated results, which can be somewhat misleading, as their reporting relies on correlation analysis — a partial approach that may lead to spurious correlations. The following sections provide formal analyses.

5.1. Results from aggregate markets

Various models associated with equation (1), which are contained in Table 3, are characterized with different restrictions. Several observations are worthy of noting. First, models 1 and 2 highlight the estimates of a negative effect of inflation on stock returns; however, the result in model 2 indicates a significant negative effect arising from the inclusion of influential dummies for the GFC and COVID-19. The evidence is consistent with the finding of multiple structural breaks in the estimation (Cifter, 2015) and negative impacts of the 2008 GFC and COVID-19 pandemic (Cheema et al., 2020; Chiang, 2022). The effect of these dummy variables can be seen in the estimated coefficients of inflation and the improvement in the adjusted R-squared.

Models 3–5 are estimated by adding the incremental variables, $\Delta emv_{inf,t}$, Δm_t and Δmpu_t , while controlling for the economic fundamental, $dy_{i,t-1}$, and dummy variables for shocks. These variables were included in order to control their correlations (whether it is positive or negative) with Inf_t , and to highlight the underlying contribution for each variable. Model 6 is considered a full model and other models are nested within it. The interpretation of this model can be summarized as follows.⁶ First, the coefficient of conditional variance is positive and statistically significant; the result is consistent with the risk-return tradeoff hypothesis discussed in the literature (Bali and Peng, 2006; Bali and Engle, 2010).

Second, it is clear that the coefficient of inflation is negative and statistically significant, supporting the notion that inflation produces an adverse effect on the economy, causing stock returns to plunge. These results are consistent with our hypothesis and agree with a common empirical regularity in industrialized economies reported by Fama (1981), Geske and Roll (1983), Marshall (1992) and

⁶ The equation excludes some recent arguments such as downside risk (Bali et al., 2009; Chen et al., 2018) and geopolitical risk (Brogaard et al., 2020; Chiang, 2021b) to avoid over parameterization.

Table 3

GED-GARCH estimates of various models for aggregate stock returns in relation to conditional variance, inflation, inflation equity market volatility, money growth, the change in monetary policy uncertainty controlling for dividend yield and the finance crisis as well as COVID-19 shocks.

Model	C	$\hat{\sigma}_t^2$	Inf_t	$\Delta emv_{inf,t}$	Δm_t	Δmpu_t	$dy_{i,t-1}$	D_{Crisis}	$D_{COVID-19}$	$\omega_{i,0}$	$\varepsilon_{i,t-1}^2$	$\sigma_{i,t-1}^2$	$\bar{R}^2$
Model 1	-3.3661	1.0376	-0.139				1.6652			10.8191	0.3593	0.6119	-0.03
	-1.59	1.99	-1.91				6.16			1.81	2.09	4.38	
Model 2	-16.1816	5.2735	-2.2018				1.5706	-10.8536	-9.3138	9.4442	0.0682	0.5651	0.06
	-3.60	3.97	-4.66				3.38	-10.14	-10.37	2.64	6.81	3.90	
Model 3	4.8867	-14.5971	-2.3588	-0.0243			1.6804	-12.4386	-3.9097	8.7457	0.1246	0.5717	0.07
	5.69	-4.64	-11.18	-25.04			6.13	-12.72	-8.03	6.01	7.87	10.10	
Model 4	-13.2345	4.2570	-0.8464	-0.0207	0.5664		1.8201	-12.2868	-17.6455	9.2210	0.1150	0.5798	0.02
	-2.95	3.26	-2.71	-12.48	4.58		5.85	-12.75	-5.26	4.28	4.26	7.90	
Model 5	-16.0971	5.3191	-1.8092	-0.0164		-0.0146	1.2930	-11.0653	-12.3332	8.9584	0.0478	0.5570	0.07
	-3.57	4.17	-11.56	-8.81		-13.07	4.00	-21.77	-6.63	4.00	5.34	5.84	
Model 6	-15.0783	4.9507	-2.5338	-0.0258	0.1793	-0.0004	2.0437	-13.1781	-8.8582	8.8017	0.1176	0.5761	0.06
	-4.51	5.63	-18.55	-15.78	1.71	-1.94	9.55	-13.80	-5.69	5.81	11.61	26.17	
Model 7	-14.7081	4.8029	-1.9912*	-0.0193	0.0774	0.0014	1.8879	-12.8984	-14.0332	8.6486	0.1055	0.5701	0.03
	-3.06	3.55	-7.04	-9.69	3.47	1.68	7.76	-13.49	-5.29	4.18	5.57	8.69	
Model 8	-11.0937	3.6474	-0.9584	-0.0165	0.2549	-0.0069	1.3320	6.1825	-7.6998	8.9122	0.1267	0.5751	0.05
	-3.69	4.63	-6.22	-7.11	3.99	-6.17	6.50	2.98	-2.75	3.62	7.53	8.86	

Notes: The estimated model is: $R_t = C + \beta_1 \hat{\sigma}_t^2 + \beta_2 Inf_t + \beta_3 \Delta emv_{inf,t} + \beta_4 \Delta m_t + \beta_5 \Delta mpu_t + \beta_6 dy_{i,t-1} + \beta_7 D_{Crisis} + \beta_8 D_{COVID-19} + \varepsilon_{i,t}$,

where the dependent variable is R_t , the aggregate stock return, which is a function of the conditional variance ($\hat{\sigma}_t^2$), inflation (inf_t), growth of money supply (m_t), change in monetary policy uncertainty (Δmpu_t), change in equity market volatility (Δmpu_t), the lagged dividend yield ($dy_{i,t-1}$), and dummy variables for models 1–6 are shifted in intercept to capture 2008 global financial crisis and COVID-19; however, the dummies in model 8 are the changes in slopes for 2008 crisis and COVID-19 expressed as respectively. Note that the estimated coefficient of inflation with an asterisk * from model 7 is based on lagged inflation.

Inf_{t-1} for IV, instead of Inf_t . The ω_0 , ε_{t-1}^2 and σ_{t-1}^2 are constant, lagged shock squared, and lagged variance for GARCH(1,1) specification. The numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of z-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

Graham (1996). Yet, this short-run result does not support the Fisher hypothesis.

Third, the coefficient of $\Delta emv_{inf,t}$ is negative and statistically significant. The negative effect is attributable to a rise in equity market volatility as it covaries with inflation, causing fears among investors and prompting a selloff of stocks. Note that the news effect of $\Delta emv_{inf,t}$ reflects real time volatility as perceived and expressed by journalists in response to the prevailing inflation. This variable differs from the direct effect of Inf_t , whose influence runs through real income or investors' money illusion. Yet, the $\Delta emv_{inf,t}$ also differs from $\hat{\sigma}_t^2$, which is derived from a model prediction conditioned on recent variance and shock squared. The evidence of a negative relation between stock returns and a change in equity market volatility is consistent with the result reported by Batten et al. (2022), who find a negative relation between stock returns and the VIX in their study of European bank returns during the COVID-19 and GFC periods, since the $emv_{inf,t}$ comoves with the VIX as constructed by (Baker et al., 2022).

This finding of $\Delta emv_{inf,t}$ is consistent with studies by Cohen et al. (2005) and Saryal (2007) that attribute the negative relation of stock returns and inflation to the risk attitude of investors. However, this study confirms that the news-based, inflation-volatility channel provides an additional force that can have a negative effect on stock returns.

Fourth, the coefficient of Δm_t is positive and significant at the 5% level. This control variable is included because "inflation is always and everywhere a monetary phenomenon"⁷ (Friedman, 1970) and the relation between stock returns and inflation is often driven by the dynamic money market conditions (Kaul, 1987). The positive coefficient is in line with the notion that expansive monetary policy increases simultaneously with stock returns as noted by Conover et al. (1999) and Ioannidis and Kontonikas (2008). This outcome also agrees with Conrad's (2021) experiments that monetary expansion has a direct positive impact on share prices. Thus, the evidence supports the liquidity hypothesis, i.e., a rise in money supply will provide more liquidity to the public and bid down the interest rate, causing an increase in stock prices as implied in the valuation theorem. This is further reinforced by the wealth effect, which indicates that a rise in money supply generates a higher level of wealth that spillovers to the stock market (Tobin, 1969).

Fifth, uncertainty can arise from a change in monetary policy, which in the context of inflation, has not been incorporated into the model as a control variable to affect stock returns. The Δmpu_t reflects journalists' forward-looking perspective on the news as it relates to policy stances, volatility, and uncertainty. A rise in MPU as reported by journalists is likely to create jitters among investors and cause a selloff and in turn stocks prices to plummet. This negative relation is shown in the estimated coefficient, which is statistically significant. The evidence is consistent with the finding documented by Chiang (2019, 2021a).

Sixth, the estimated coefficient for the lagged dividend yield exhibits a positive sign and is highly significant. This result is consistent with the valuation model (Campbell et al. 1997; Fama and French, 1988) that holds a reported higher dividend heralds a rise in future cash flows and leads to higher stock returns. The evidence is also in line with the studies by Bali et al. (2009) and Chen et al. (2018).⁸

Using the current inflation rate as an independent variable in model estimation may lead to the issue of simultaneity. The effect of simultaneity may arise because of a time delay in CPI index announcement which cause a lagged response to stock returns or a delayed reaction to news due to real options (Pindyck, 1991), which can be another concern. To address these issues, we employ the lagged inflation as an instrumental variable (IV) in the estimated equation, labelled model 7. The estimated coefficient on lagged inflation is -1.9912 , which is significant at the 1% level, while the other coefficients are qualitatively comparable. Thus, the model is robust across different measures of inflation rate.

In the estimations of models 2–7, the impacts of the 2008 GFC or COVID-19 shock are assumed to be mainly on the intercept. It is perceivable that the impacts can affect the slope due to these external shocks. The impacts are captured by the terms of $(D_{Crisis} \bullet Inf_t)$ and $(D_{COVID-19} \bullet Inf_t)$ for the GFC and COVID-19 pandemic, respectively. Estimates using this specification are reported in model 8 of Table 3. Evidence shows that coefficients of inflation continually exhibit negative signs and are statistically significant. However, the coefficients of the slope estimations exhibit diverse effects, indicating that investors reacted to the shocks from the GFC and COVID-19 in different ways.

5.2. Result from sectoral markets

5.2.1. Linear measures of inflation

Two significant empirical results are derived from the aggregate market estimations. First, the evidence allows us to conclude that stock returns and inflation are negatively correlated, which suggests stocks cannot be used to hedge against inflation risk. Second, stock returns are negatively correlated with equity market volatility attributable to inflation. This channel provides an additional pathway for understanding how inflation affects stock returns. Yet, for investors who form portfolios based on sectoral stocks, there is no convincing reason to believe that the above results hold true. To test the ability of stocks to hedge against inflation using sectoral stocks, we write

⁷ Milton Friedman (1970) famously stated in full: "Inflation is always and everywhere a monetary phenomenon, in the sense that it is and can be produced only by a more rapid increase in the quantity of money than in output."

⁸ To gain further insights of the estimated results, we extend the data to April 2023 and incorporate data from the G7 countries. The estimates for model 6 are reported in the table of Appendix. In general, the evidence decisively supports that inflation has a strong negative effect on stock returns, and the uncertainty from the US market will also produce an adverse effect. The evidence illustrates that US monetary policy or equity market volatility create not only its own market downturn, but also spillover to the other advanced markets. Among all estimated coefficients, only the coefficients of monetary growth for CA, GM, and IT have diverse signs, which may arise from a distinctive monetary policy stance, market reactions, or simply a spurious correlation.

Table 4

Estimates of sectoral stock returns in relation to the conditional variance and inflation equity market volatility while controlling for dividend yields, finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{i,t}^2$	Inf_t	$\Delta emv_{inf,t}$	$dy_{i,t-1}$	$\omega_{i,0}$	$\epsilon_{i,t-1}^2$	$\sigma_{i,t-1}^2$	R^2
FINL	-2.1150	0.5570	-2.0423	-0.0227	3.2599	1.6686	0.1923	0.8068	0.02
	-1.73	1.88	-3.87	-5.11	3.75	1.26	2.15	10.32	
HLCA	-0.6992	0.1564	0.0664	-0.0090	2.9238	9.4886	0.3673	0.6062	0.01
	-0.59	0.48	0.45	-4.47	6.36	0.75	0.87	1.51	
INDU	-1.9573	0.7218	-1.0755	-0.0141	0.6504	7.7638	0.3879	0.5644	0.01
	1.34	0.36	-0.47	0.00	0.18	4.90	0.22	0.18	
REIT	1.0336	-0.9137	-1.4224	-0.0135	2.2563	4.1079	0.2510	0.6934	0.01
	0.59	-2.06	-2.61	-3.11	3.99	1.70	1.92	5.56	
CNSG	-4.7004	1.1165	-0.6255	-0.0239	2.4001	15.7000	0.2055	0.6071	0.01
	-1.22	1.16	-2.99	-7.67	4.55	1.41	1.06	2.75	
RTAL	1.2748	0.0450	-1.2584	-0.0231	1.0860	19.0104	0.2962	0.6511	0.04
	9.44	1.12	-2.53	-6.39	4.65	1.96	0.84	3.15	
BMAT	-0.1381	-0.4545	2.7877	-0.0305	3.1099	18.1517	0.5687	0.5430	0.04
	-0.35	-3.68	6.51	-8.20	7.03	1.16	1.49	1.88	
ENGY	-3.8818	0.9290	2.1256	-0.0255	-0.0026	27.4120	0.2035	0.6505	0.04
	-0.88	0.98	5.46	-6.88	-0.01	4.07	0.99	4.82	
INVS	-3.6233	0.9531	2.1255	-0.0308	2.2464	15.5125	0.7227	0.6536	0.01
	-1.69	2.12	5.03	-7.51	6.26	1.70	2.10	9.20	
TECH	0.0016	0.3393	1.7433	-0.0369	0.7423	41.3105	0.7168	0.6456	0.03
	0.03	11.75	4.45	-17.58	5.93	1.01	1.04	2.30	

Notes: The dependent variable of sectoral stock at time t is $R_{i,t}$, i refers to sectors, including Financials (FINL), Health cares (HLCA), Industrials (INDU), Real estate (REIT), Consumption goods (CNSG), Retailers (RTAL), Basic Materials (BMAT), Energy (ENGY), Investment services (INVS), and Technology (TECH). $\hat{\sigma}_{i,t}^2$ is the conditional variance projected by a GED-GARCH(1,1) process. The inf_t is inflation, $\Delta emv_{inf,t}$ is the equity market volatility attributable to inflation, $dy_{i,t-1}$ is the lagged dividend yield in natural log in sector. The ω_0 , $\epsilon_{i,t-1}^2$ and $\sigma_{i,t-1}^2$ are constant, lagged shock squared, and lagged variance, respectively, for the GARCH(1,1) specification. The numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of z-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the R-squared.

$$R_{i,t} = C + \beta_1 \hat{\sigma}_{i,t}^2 + \beta_2 Inf_t + \beta_3 \Delta emv_{inf,t} + \beta_4 \Delta m_t + \beta_5 \Delta mpu_t + \beta_6 dy_{i,t-1} + \beta_7 D_{Crisis} + \beta_8 D_{COVID-19} + \epsilon_{i,t} \quad (5)$$

Equation (5), which possesses nearly the same properties as those in equation (1); the primary difference is that the variables $\{R_{i,t}, \hat{\sigma}_{i,t}^2, dy_{i,t-1}, \epsilon_{i,t}\}$ with subscript i refer to individual sector stock i . Other variables are defined as before.

The estimated results of equation (5) using a GED-GARCH(1,1) procedure are reported in Tables 4 and 5. Results in Table 4 are obtained by restricting $\beta_4 = \beta_5 = \beta_7 = \beta_8 = 0$, but this restriction is relaxed in the estimates reported in Table 5. The significance of Δm_t , Δmpu_t and two dummy variables D_{Crisis} and $D_{COVID-19}$ in Table 5 validates the inclusion of these variables and highlights the impacts of parametric bias when these variables are excluded in Table 4.

From Table 5, it is evident that the estimated coefficients of inflation show that six out of 10 sectors display negative signs and are statistically significant. The evidence is in line with the performance from the aggregate data and is consistent with the literature (Fama and Schwert, 1977; Fama, 1981). However, the coefficients of inflation for the BMAT, ENGY, INVS and TECH sectors are positive and statistically significant (except for BMAT). The test results in Table 5 provide solid evidence that the ENGY, INVS and TECH sectors have the ability to hedge against inflation risk. This finding is partly consistent with Batten et al. (2021), who distinct benefits from hedging stocks with oil.

Evidence in Table 5 also indicates another channel by which inflation can produce a negative impact on stock returns, i.e., via its impact on equity market volatility. This is seen from the negative coefficients of $\Delta emv_{inf,t}$ which are highly significant. Note that the negative sign: $-\Delta emv_{inf,t} = -emv_{inf,t} + emv_{inf,t-1}$ implies that investors who bought stocks at time $(t-1)$ during a period of high inflation-induced volatility will be rewarded by a risk premium at time t as inflation subsidies and volatility moderates. Attributable to inflation, this risk premium differs from that due to conditional volatility, $\hat{\sigma}_{i,t}^2$, which is generated from the stock market volatility *per se*. It should be mentioned that the coefficient of the BMAT sector exhibits a significantly positive sign for $\Delta emv_{inf,t}$, indicating that the BMAT stocks have the ability to hedge against inflation. However, ENGY and other sectors that bear negative signs do not exhibit an ability to hedge against market volatility. The evidence is consistent with the results documented by Hassan et al. (2022), who find that crude oil could not offer a safe haven against Gulf Cooperation Council (GCC) sectoral equity markets. Their study concludes that returns of GCC equity markets are vulnerable to volatility and risk in the global financial markets.

The coefficients for the remaining variables are also consistent with the results from the aggregate data. For instance, the coefficients of Δm_t are positive and statistically significant; these outcomes are attributable to either a liquidity effect following money growth or a wealth effect due to an increase in wealth that spills over to a demand for stocks. Next, the coefficients of Δmpu_t are negative and statistically significant. Here we find another policy uncertainty premium due to monetary policy uncertainty, which reveals as a negative sign at time t ($-mpu_t$) but a positive sign at time $t-1$ (mpu_{t-1}). This result indicates a reward is granted to investors

Table 5

Estimates of the stock returns in relation to the conditional variance, inflation, inflation equity market volatility, money growth, monetary policy uncertainty while for controlling for dividend yield, finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{it}^2$	Inf_t	$\Delta emv_{inf,t}$	Δm_t	Δmpu_t	dy_{t-1}	D_{Crisis}	$D_{COVID-19}$	$\omega_{i,0}$	ϵ_{it-1}^2	σ_{it-1}^2	$\bar{R}^2$
TTMK	-15.0783	4.9507	-2.5338	-0.0258	0.1793	-0.0004	2.0437	-13.1781	-8.8582	8.8017	0.1176	0.5761	0.06
	-4.51	5.63	-18.55	-15.78	1.71	-1.94	9.55	-13.80	-5.69	5.81	11.61	26.17	
FINL	-8.3291	2.4803	-1.6909	-0.0051	0.0908	-0.0202	1.2406	-4.6595	-12.8588	16.2497	0.2127	0.5760	0.04
	-1.47	1.78	-4.47	-3.14	12.42	-12.30	2.80	-1.73	-3.88	2.21	1.81	4.56	
HLCA	-7.4088	2.2365	-1.2478	-0.0064	0.6767	-0.0097	3.2823	-6.0926	-6.9282	9.6385	0.0699	0.5895	0.03
	-1.01	0.98	-3.76	-2.62	4.82	-5.68	8.38	-7.09	-7.40	3.48	1.00	3.96	
INDU	-10.0069	2.9252	-1.0118	-0.0144	0.5930	-0.0091	2.2243	-14.6795	-9.5889	14.1646	0.1184	0.5838	0.07
	-2.65	3.23	-5.38	-6.23	5.92	-8.81	6.66	-12.72	-3.09	3.41	3.76	8.05	
REIT	-2.6092	-0.2263	-0.5280	-0.0181	0.4708	-0.0080	3.0895	-0.5010	-6.6489	8.2388	0.5331	0.6182	0.01
	-2.26	-0.90	-2.08	-6.86	4.14	-4.98	8.37	-0.29	-1.85	1.76	1.35	3.68	
CNSG	-6.8261	1.4263	-2.1685	-0.0072	0.8304	-0.0171	3.5764	-8.9605	-9.6196	13.8818	0.1845	0.6024	0.09
	-21.91	14.33	-8.25	-10.13	7.68	-14.40	8.70	-9.59	-2.87	2.95	6.87	5.08	
RTAL	-0.4502	0.4453	-1.5083	-0.0240	0.9908	-0.0137	1.5922	-3.6495	-5.5130	29.1666	0.1695	0.5963	0.05
	-0.59	2.59	-3.29	-6.87	7.36	-9.82	5.76	-5.53	-4.38	0.42	1.86	0.70	
BMAT	-1.6814	-0.2097	0.0301	1.1580	-0.0131	-0.0251	3.5513	-10.2276	-7.6304	1.9858	0.1202	0.8169	0.11
	-0.68	-0.27	0.03	2.75	-2.54	-3.29	2.14	-5.17	-2.58	1.62	2.80	13.57	
ENGY	-6.3993	1.4733	0.8227	-0.0113	0.6699	-0.0196	0.4600	-4.5647	-15.997	26.9674	0.2329	0.6674	0.08
	-1.27	1.38	2.37	-3.86	4.79	-18.87	0.92	-3.35	-61.43	1.36	3.21	3.93	
INVS	-8.1604	1.9194	0.1835	-0.0229	0.2524	-0.0279	2.6508	-2.5015	-12.2760	40.1772	0.246	0.6013	0.04
	-9.19	10.67	2.68	-6.59	5.26	-9.75	6.59	-0.82	-4.93	2.60	2.59	4.55	
TECH	-0.2079	0.3556	0.7648	-0.0336	0.7815	-0.0184	1.0604	-4.3083	-7.8646	29.9462	0.1938	0.7012	0.05
	-0.84	7.35	4.92	-8.87	4.89	-7.87	6.93	-2.36	-4.48	0.73	1.04	2.05	

Notes: The figures in the first two rows are associated with the aggregate market and contain no subscribe i for the related variables. The dependent variable of the aggregate stock return (TTMK) at time t is R_t or sector i , R_{it} . The $\hat{\sigma}_{it}^2$ is the conditional variance projected by a GED-GARCH(1,1) process; Inf_t is inflation; $\Delta emv_{inf,t}$ is the US equity market volatility attributable to inflation risk; Δm_t is money growth; Δmpu_t is the change in monetary policy uncertainty; dy_{t-1} is the lagged dividend yield in the natural log in sector i ; D_{Crisis} and $D_{COVID-19}$ are the dummy variables. The ω_0 , ϵ_{it-1}^2 and σ_{it-1}^2 are constant, lagged shock squared, and lagged variance, respectively, for the GARCH(1,1) specification. The numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

for purchasing stock during a high uncertainty period as noted by Chatziantoniou et al. (2013), Husted et al. (2020) and Chiang (2020). Third, coefficients of $dy_{i,t-1}$ are positive and statistically significant. This finding is consistent with the textbook notion of the dividend valuation theorem in asset pricing (Campbell et al., 1997, Chen et al., 2018). Finally, the dummy variables of D_{Crisis} and $D_{COVID-19}$ are negative and statistically significant. The negative signs reflect the stock market downturns, a response to a liquidity crisis that caused business failures in 2008 and also to COVID-19, which in turn triggered major interruptions of supply chains or business shutdowns and had a devastating impact on economies as a worsening pandemic threaded itself around the world (Chiang, 2022). As noted by Tsay (1988) and Pena (2001), without controlling for these influential observations, the estimated parameters are doomed to have estimated biasness.

Chen et al. (1986), Amihud (1996) and Al-khazali and Pyun (2004) suggest that both expected and unexpected inflation can be alternatively used to examine the relation between inflation and stock returns. Based on this line of reasoning, the inflation series in the US market is identified as an AR(1), AR(2) and AR(12) process, which follows the Box-Jenkin's methodology (Pena et al., 2001).⁹ Table 6 reports the estimates by replacing Inf_t with expected inflation Inf_t^e . The latter is obtained using the optimal forecast of Inf_t for an ARIMA method.¹⁰ A review of the estimated coefficients of Inf_t^e shows that eight out of 10 sectors show negative signs, which is consistent with conventional results. However, the BMAT and ENGY sectors are positive and highly significant, indicating that these two sectoral stocks can have the ability to hedge against inflation. The findings seem in line with that of Bouri et al. (2023). However, since the coefficients of $\Delta emv_{inf,t}$ continually show negative signs, the results indicate these sectors still do not have the unequivocal ability to hedge against stock market volatility driven by inflation risk. However, the magnitude of coefficients from $\Delta emv_{inf,t}$ are relatively smaller as compared with that of Inf_t^e ; therefore, the net effect for the BMAT and ENGY sectors is still positive, indicating a continuing ability to hedge against inflation.

Table 7 reports the estimated result using unexpected inflation, Inf_t^u , in the test equation, where $Inf_t^u = Inf_t - Inf_t^e$ (Mishkin, 1983; Cornell, 1983). The test results yield additional sectors with positive coefficients; in total, they include INDU, REIT, BMAT, ENGY and TECH. However, the coefficients of $\Delta EMV_{inf,t}$ are significantly negative although the values are relatively small. In sum, the evidence from Tables 5-7 suggests that the stocks in sectors BMAT and ENGY consistently demonstrate positive signs and are able to hedge inflation. The evidence is consistent with the finding by Bouri et al. (2023), who tested U.S. sectoral data in their wavelet analysis and found that energy and basic material are pro-inflation sectors. However, Hassan et al. (2022) find that crude oil could not offer safety for GCC's sectoral equity markets. Further, their results show that returns of GCC equity markets are vulnerable to volatility and risk in the global financial markets.

5.2.2. Different regimes of inflation changes

It is perceivable that market participants may respond differently to a rise in inflation vis-à-vis a decline (Ozoguz, 2009; Lee, 2010). To test the directional impact, we divide a change in inflation into two regimes: A rise in inflation, $\Delta Inf_t^+ = \Delta Inf_t > 0$, as bad news; a decline in inflation, $\Delta Inf_t^- = \Delta Inf_t < 0$, as good news. Accordingly, the estimated equation can be expressed as:

$$R_{i,t} = C + \beta_1 \sigma_{i,t}^2 + \beta_2 \Delta Inf_t^+ + \beta_3 \Delta Inf_t^- + \beta_4 emv_{inf,t} + \beta_5 \Delta m_t + \beta_6 \Delta mp_u + \beta_7 dy_{i,t-1} + \beta_8 D_{Crisis} + \beta_9 D_{COVID-19} + \varepsilon_{i,t} \quad (6)$$

Estimates of equation (6) using the GED-GARCH procedure are reported in Table 8.

Evidence clearly indicates that bad news about inflation causes a downturn in stock returns, as reflected by the negative and significant coefficients of ΔInf_t^+ . The only exception is the ENGY sector, which shows a positive sign and is highly significant, suggesting that only the ENGY stocks can be used to hedge against inflation risk. This property is further supported by the positive and significant sign of the coefficient of Δemv_{inf} for ENGY stocks.

The estimated results also show that a decline in inflation, which is viewed as good news, would bring about a positive effect on stock returns. This is shown in the positively significant coefficients for ΔInf_t^- . Thus, splitting the change in inflation into two different regimes provides much clearer and convincing evidence for explaining the stock-inflation relation. As in the previous estimations, the coefficients of the $\Delta emv_{inf,t}$ consistently show negative signs, reflecting investors' negative response to equity market volatility caused by concerns about the inflationary impact of equity market risk. It is evident that results from Table 8 unambiguously allow us to conclude that a rise in inflation will produce damaging effects on stock returns due to both the direct effect of real money balance and the market-volatility effect triggered by a higher inflation.

With respect to the other variables, the coefficients of $\{\sigma_{i,t}^2, \Delta m_t, dy_{i,t-1}\}$ with two minor exceptions have a positive effect on stock returns, while Δmp_u has a negative effect on stock returns. These results are comparable to those in the previous estimations, which we shall not repeat in the discussion.

⁹ Some researchers (Geske and Roll, 1983; Beladi et al., 1993; Chiang, 2023) prefer to use adaptive expectations to capture an error learning behavior of the inflation evolution.

¹⁰ The ARIMA equation is: $Inf_t = 0.208 + 0.575 Inf_{t-1} - 0.153 Inf_{t-2} - 0.171 Inf_{t-12} + \varepsilon_t$, $\bar{R}^2 = 0.27$, $Q(24) = 26.90$, t-statistics: (13.06) (11.47) (-3.01) (-3.60).

Table 6

Estimates of stock returns in relation to conditional variance, expected inflation, inflation equity market volatility, money growth, monetary policy uncertainty while controlling for dividend yield, dummies for finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{it}^2$	Inf_t^e	$\Delta emv_{inf,t}$	Δm_t	Δmpu_t	Δdy_{it-1}	D_{Crisis}	$D_{COVID-19}$	ω_{i0}	ε_{it-1}^2	σ_{it-1}^2	$\bar{R}^2$
TTMK	-15.8348	5.3134	-6.3072	-0.0265	0.9652	-0.0067	2.6839	-7.4008	-7.4873	7.9447	0.1186	0.5609	0.14
	-2.87	3.09	-2.30	-9.68	6.88	-4.12	7.26	-16.26	-3.99	4.30	2.55	10.73	
FINL	-6.7534	2.7007	-16.6955	-0.0074	0.2126	-0.0200	2.2313	-6.8795	-12.3432	15.2290	0.1506	0.5700	0.10
	-1.94	3.58	-3.16	-2.58	3.24	-14.77	5.50	-3.86	-5.57	3.91	2.78	5.07	
HLCA	-4.5558	1.8702	-11.0126	-0.0115	0.6926	-0.0078	3.3912	-5.7939	-5.6308	9.4612	0.1488	0.5850	0.05
	-2.32	3.78	-3.26	-4.51	5.68	-5.25	9.55	-5.48	-19.40	1.98	2.60	3.40	
INDU	-11.9228	3.4544	-4.0836	-0.0107	0.8147	-0.0123	2.8161	-12.7835	-8.6716	13.2134	0.1219	0.5704	0.12
	-2.38	2.46	-1.98	-4.59	7.77	-8.84	9.39	-12.57	-4.09	3.24	2.23	7.48	
REIT	-0.5699	-0.0555	-13.8379	-0.0099	0.7002	-0.0098	2.9894	-1.2087	-6.4390	2.9070	0.2144	0.8117	0.03
	-2.02	-0.51	-15.06	-3.00	4.70	-6.23	14.37	-0.57	-1.96	0.82	1.25	5.43	
CNSG	-9.5716	2.4101	-3.9118	-0.0126	0.9533	-0.0178	3.7916	-8.8382	-9.6569	13.0454	0.1250	0.5908	0.05
	-7.90	5.77	-2.09	-6.62	8.27	-16.89	10.31	-9.23	-3.09	2.02	3.14	3.65	
RTAL	-1.1842	1.2781	-11.1544	-0.0217	0.9739	-0.0117	1.7027	-3.5706	-5.9883	15.8799	0.2235	0.5972	0.04
	-3.78	5.72	-2.05	-11.56	11.04	-8.73	7.19	-3.94	-4.25	2.70	2.62	5.21	
BMAT	-15.7421	2.5856	5.5140	-0.0238	0.9855	-0.0112	6.3602	-13.7838	-8.5846	18.0819	0.1229	0.5739	0.14
	-5.28	3.40	2.77	-6.09	7.15	-5.44	10.48	-7.77	-2.45	2.01	2.46	3.26	
ENGY	-4.5171	1.1153	2.6966	-0.0107	0.9777	-0.0088	0.1198	-9.9792	-15.2820	22.9240	0.1405	0.6167	0.09
	-0.78	0.82	6.77	-2.37	6.88	-4.73	2.39	-6.07	-2.37	1.54	1.99	2.92	
INVS	-4.1809	2.6720	-33.5601	-0.0213	0.5791	-0.0275	4.3006	-10.7169	-13.7233	40.4198	0.2380	0.6070	0.03
	-0.97	3.26	-15.41	-5.55	4.91	-11.53	9.40	-3.84	-3.64	2.05	6.82	4.72	
TECH	-0.9723	0.8258	-3.5598	-0.0360	0.7698	-0.0155	1.2382	-4.9632	-7.2670	28.2621	0.3572	0.5944	0.04
	-0.40	2.79	-0.65	-11.67	4.21	-6.96	6.87	-3.01	-3.37	1.28	3.06	2.85	

Notes: The figures in the first two rows are associated with the aggregate market and contain no subscribe i for the related variables. The dependent variable of the aggregate stock return (TTMK) at time t is R_t or sector i, R_{it} . The $\hat{\sigma}_{it}^2$ is the conditional variance projected by a GED-GARCH(1,1) process; Inf_t^e is the expected inflation; $\Delta emv_{inf,t}$ is the US equity market volatility attributable to the inflation risk; Δm_t is money growth; Δmpu_t is a change in monetary policy uncertainty; dy_{it-1} is the lagged dividend yield in natural log in sector i; D_{Crisis} and $D_{COVID-19}$ are the dummy variables. The ω_0 , ε_{t-1}^2 and σ_{t-1}^2 are constant, lagged shock squared, and lagged variance for GARCH(1,1) specification. Numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

Table 7

Estimates of stock returns in relation to conditional variance, unexpected inflation, inflation equity market volatility, money growth, monetary policy uncertainty while controlling for dividend yield, dummies variables for finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{it}^2$	Inf_t^u	$\Delta emv_{inf,t}$	Δm_t	$\Delta mp u_t$	dy_{it-1}	D_{Crisis}	$D_{COVID-19}$	$\omega_{i,0}$	ε_{it-1}^2	σ_{it-1}^2	$\bar{R}^2$
TTMK	-17.9072	5.6652	-0.3522	-0.0113	0.2516	-0.0065	2.5912	-9.5814	-8.8353	8.4636	0.1021	0.5743	0.11
	-8.20	7.98	-3.94	-5.08	3.04	-6.03	7.40	-8.99	-4.44	10.75	5.80	9.84	
FINL	-10.1817	2.6290	-0.6597	-0.0110	0.2154	-0.0176	2.5826	-8.0916	-12.4386	15.0422	0.1763	0.5684	0.10
	-6.34	5.17	-2.74	-3.98	7.78	-10.52	5.10	-3.51	-3.99	2.92	4.31	5.08	
HLCA	-6.3034	1.7411	-0.2589	-0.0135	0.5841	-0.0062	3.6810	-5.9244	-5.6307	9.2855	0.1149	0.5861	0.04
	-6.00	4.53	-2.31	-5.10	5.33	-4.46	10.59	-5.94	-3.63	2.98	3.67	4.49	
INDU	-12.5454	3.4476	0.6418	-0.0223	0.5726	-0.0130	2.7779	-12.0093	-8.7251	13.6900	0.0854	0.5806	0.12
	-1.85	1.88	4.32	-10.76	5.22	-15.89	9.96	-14.67	-2.91	3.00	2.31	4.36	
REIT	-4.2320	0.1150	0.4201	-0.0126	0.6784	-0.0113	3.1402	0.1336	-6.3045	12.7841	0.3327	0.6003	0.01
	-7.48	2.14	1.97	-4.50	6.01	-9.58	10.94	0.33	-1.83	0.91	0.92	1.64	
CNSG	-10.2473	2.2730	-0.8624	-0.0099	0.6931	-0.0131	3.5469	-9.0710	-9.1515	12.3464	0.1676	0.5760	0.10
	-3.40	2.89	-3.57	-5.89	5.29	-11.76	7.72	-9.80	-2.99	2.93	5.66	4.93	
RTAL	-3.5202	1.3169	-0.3236	-0.0229	1.1024	-0.0130	2.0638	-3.7956	-5.7567	16.0954	0.1521	0.6000	0.03
	-3.81	5.81	-6.20	-8.17	8.62	-11.26	8.20	-5.66	-6.38	2.60	2.55	3.67	
BMAT	-15.5179	2.9303	1.4952	-0.0156	0.7159	-0.0084	6.4728	-14.2261	-9.3916	18.2985	0.1346	0.5683	0.12
	-6.28	4.01	3.05	-5.21	6.61	-4.46	12.46	-9.17	-2.77	3.01	5.38	4.54	
ENGY	-6.4113	1.3947	3.5144	-0.0262	0.9819	-0.0108	-0.5621	-3.8438	-14.2627	24.6531	0.3881	0.6631	0.08
	-1.41	1.47	30.16	-13.12	10.31	-15.39	-3.94	-4.09	-1.87	1.38	2.23	4.86	
INVS	-4.1809	2.6720	-33.560	-0.0213	0.5791	-0.0275	4.3006	-10.7169	-13.7233	40.4198	0.2380	0.6070	0.03
	-0.97	3.26	-15.41	-5.55	4.91	-11.53	9.40	-3.84	-3.64	2.05	6.82	4.72	
TECH	-2.5700	1.0101	1.2109	-0.0342	0.6404	-0.0147	1.6073	-5.2643	-7.2143	28.5934	0.3514	0.5961	0.05
	-3.46	7.06	3.48	-10.89	5.28	-6.80	9.03	-3.25	-6.03	4.63	1.78	5.01	

Notes: The figures in the first two row are associated with the aggregate market and contain no subscribe i for the related variables. The dependent variable of the aggregate stock return (TTMK) at time t is R_t or sector i, R_{it} . The $\hat{\sigma}_{it}^2$ is the conditional variance predicted by a GED-GARCH(1,1) process; Inf_t^u is the unexpected inflation; $\Delta emv_{inf,t}$ is the US equity market volatility attributable to the inflation risk; Δm_t is money growth; $\Delta mp u_t$ is a change in monetary policy uncertainty; dy_{it-1} is the lagged dividend yield in natural log in sector i; D_{Crisis} and $D_{COVID-19}$ are the dummy variables. The ω_0 , ε_{t-1}^2 and σ_{t-1}^2 are constant, lagged shock squared, and lagged variance for GARCH(1,1) specification. Numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

Table 8

Estimates of stock returns in relation to conditional variance, inflation rise, inflation down, inflation equity market volatility, money growth, monetary policy uncertainty while for controlling dividend yield, dummies for finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{it}^2$	ΔInf_t^+	ΔInf_t^-	Δemv_{inf}	Δm_t	Δmpu_t	dy_{it-1}	D_{Crisis}	$D_{COVID-19}$	$\omega_{i,0}$	ε_{it-1}^2	σ_{it-1}^2	$\bar{R}^2$
TTMK	-18.9370	6.3887	-3.3754	5.1661	-0.0244	0.2856	-0.0031	1.3373	-8.3096	-8.9622	8.7931	0.1003	0.5741	0.06
	-4.98	6.24	-5.56	8.83	-8.18	2.30	-2.75	3.53	-9.98	-15.36	4.97	4.81	10.09	
FINL	-9.8113	2.9444	-2.4170	3.3695	-0.0117	0.0114	-0.0197	1.2699	-3.0578	-12.7944	15.4382	0.1303	0.5674	0.07
	-2.21	2.56	-4.95	7.16	-4.48	1.83	-32.63	2.93	-1.86	-4.38	4.20	3.52	5.14	
HLCA	-7.8750	2.2444	-0.2227	0.5539	-0.0084	0.7234	-0.0101	3.1889	-5.9947	-5.7085	9.5973	0.1287	0.5774	0.03
	-8.11	7.54	-4.38	2.90	-4.40	6.24	-6.54	12.20	-5.54	-3.62	2.77	4.05	4.01	
INDU	-10.6265	3.0779	-1.6572	2.6154	-0.0194	0.8834	-0.0094	2.2075	-12.4456	-8.9594	13.5732	0.1179	0.5561	0.10
	-1.85	2.01	-3.40	4.91	-7.88	6.49	-6.03	6.16	-8.53	-3.11	2.15	2.63	3.52	
REIT	-3.3102	-0.0221	-1.4077	0.6114	-0.0175	0.7229	-0.0086	3.1552	-1.4161	-6.6511	2.9424	0.2279	0.8115	0.01
	-6.71	-0.92	-2.22	2.13	-6.77	6.32	-4.51	11.56	-0.60	-2.12	0.87	1.27	5.81	
CNSG	-3.7104	0.7162	-1.0996	1.5185	-0.0147	1.2801	-0.0078	2.3617	-6.0566	-7.7833	13.2472	0.2404	0.6008	0.08
	-1.75	1.31	-2.84	3.24	-5.71	9.51	-6.27	6.19	-5.52	-2.21	1.77	2.32	3.17	
RTAL	-1.1055	0.5720	-1.5475	0.1077	-0.0256	1.0599	-0.0139	1.5833	-2.4497	-5.7131	16.8498	0.1895	0.6373	0.05
	-2.09	4.56	-3.32	1.01	-7.33	7.87	-6.45	6.54	-2.15	-3.16	8.04	4.04	9.12	
BMAT	-11.1686	2.1641	-1.3826	3.5780	-0.0281	1.2011	-0.0036	4.2196	-11.4215	-8.8414	19.9703	0.2307	0.5937	0.10
	-12.11	8.15	-7.17	15.37	-9.85	16.90	-6.40	10.64	-15.24	-2.62	4.36	5.84	5.79	
ENGY	-4.1019	1.3169	0.890	5.4191	1.0004	-0.0152	-0.0286	0.2982	-10.7862	-16.1400	22.5531	0.1626	0.6049	0.08
	-1.62	2.41	6.65	14.47	9.70	-17.52	-15.65	10.51	-12.76	-2.23	1.89	3.42	3.94	
INVS	-9.9289	2.5370	-1.4285	4.2717	-0.0235	0.7147	-0.0258	2.5829	-7.6500	-13.1481	40.9051	0.2561	0.6040	0.02
	-3.20	4.26	-2.62	5.04	-4.85	3.50	-8.69	5.17	-2.84	-3.48	2.81	2.13	5.35	
TECH	0.1879	0.3505	-0.0210	2.5145	-0.0344	0.6785	-0.0183	1.1140	-4.4588	-7.2585	53.9628	0.4790	0.5491	0.05
	0.60	5.40	-2.59	6.98	-14.34	5.34	-11.33	7.35	-2.81	-7.13	1.28	1.81	2.49	

Notes: The dependent variable of the aggregate stock return (TTMK) at time t is R_t or sector i , $R_{i,t}$. The $\hat{\sigma}_{it}^2$ is conditional variance projected by a GED-GARCH(1,1) process; $\Delta Inf_t^+ = \Delta Inf_t > 0$ which is a rise in inflation; while $\Delta Inf_t^- = \Delta Inf_t < 0$, a decline in inflation; Δm_t is money growth; $\Delta emv_{inf,t}$ is the US equity market volatility attributable to inflation risk; Δmpu_t is a change in monetary policy uncertainty; dy_{it-1} is the lagged dividend yield in natural log in sector i ; D_{Crisis} is the dummy variable, which is set to unity for the 2008 world finance crisis and zero otherwise; $D_{COVID-19}$ is the dummy variable, which is set to unity for the COVID-19 period and zero otherwise. Numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

Table 9

Estimates of stock returns in relation to conditional variance, inflation rise, lagged inflation rise, inflation down, lagged inflation down, inflation equity market volatility, money growth, monetary policy uncertainty while controlling for dividend yield and dummies variables for finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{it}^2$	ΔInf_t^+	ΔInf_{t-1}^+	ΔInf_t^-	ΔInf_{t-1}^-	Δemv_{inf}	Δm_t	Δmpu_t	$dy_{i,t-1}$	D_{Crisis}	$D_{COVID-19}$	$\bar{R}^2$
TTMK	-17.3027	5.7828	-0.8562	-3.3822	1.7080	2.6302	-0.0032	0.3089	-0.0159	1.8015	-6.9551	-9.1477	0.09
	-3.83	4.27	-2.21	-5.85	3.57	7.26	-2.80	2.99	-8.06	5.35	-6.52	-36.35	
FINL	-10.8819	3.2827	-2.9081	-3.8439	2.1167	-1.1311	-0.0217	0.0169	-0.0105	1.4174	-4.2404	-13.032	0.10
	-2.39	2.79	-2.98	-5.71	2.57	-2.03	-10.37	0.30	-3.01	2.49	-2.21	-4.45	
HLCA	-7.6874	2.1978	0.4277	-1.6557	0.6154	0.3094	-0.0067	0.5416	-0.0094	3.5442	-6.2271	-5.4958	0.04
	-2.34	2.12	1.94	-6.92	3.41	4.74	-4.47	4.76	-4.45	9.53	-5.34	-3.10	
INDU	-10.8250	3.1796	-1.0870	-1.8005	1.9848	0.3040	-0.0118	0.6261	-0.0197	2.1975	-11.980	-9.2288	0.10
	-2.57	2.85	-2.23	-3.80	5.67	2.15	-6.94	4.11	-5.93	5.53	-9.10	-3.21	
REIT	-3.5049	0.1066	-0.3776	-2.4991	1.1148	-2.9119	-0.0114	0.4866	-0.0181	2.8288	-1.7964	-6.6786	0.04
	-3.08	0.47	-2.11	-5.80	3.82	-6.32	-9.03	4.37	-8.74	11.94	-0.85	-1.64	
CNSG	-3.8738	0.7677	-1.0913	-0.6673	-0.5561	1.1580	-0.0144	1.2384	-0.0099	2.4998	-8.4542	-8.1232	0.08
	-2.14	1.81	-2.96	-3.07	-1.28	3.80	-9.96	10.35	-4.31	9.06	-6.90	-36.47	
RTAL	-1.1788	0.5413	-1.4161	-0.0004	0.2549	-2.1032	-0.0136	1.0277	-0.0265	1.7733	-3.5235	-4.4087	0.04
	-0.62	1.32	-3.09	-0.03	2.80	-9.52	-11.48	13.58	-13.11	7.94	-3.67	-2.94	
BMAT	-11.1677	2.1956	-1.3915	-2.9798	3.0739	-0.2539	-0.0136	1.0418	-0.0304	4.7324	-8.5317	-10.466	0.09
	-2.31	1.77	-4.41	-5.98	7.63	-3.18	-6.96	6.44	-8.32	9.29	-5.45	-6.05	
ENGY	1.2807	-4.2764	1.5713	-0.8340	6.8039	2.2971	-0.0078	0.8348	-0.0246	0.1151	-6.1463	-16.087	0.08
	1.94	-1.42	2.40	-1.30	14.56	3.61	-5.45	5.49	-7.70	1.65	-5.18	-2.26	
INVS	-10.8250	3.1796	-1.0870	-1.8005	1.9848	0.3040	-0.0118	0.6261	-0.0197	2.1975	-11.980	-9.2288	0.10
	-2.57	2.85	-2.23	-3.80	5.67	2.15	-6.94	4.11	-5.93	5.53	-9.10	-3.21	
TECH	0.0208	0.4596	0.6428	-1.8066	1.9637	2.3839	-0.0171	0.6009	-0.0342	1.1001	-3.1048	-7.1855	0.04
	2.20	14.83	1.81	-15.02	7.71	6.68	-11.12	4.49	-19.32	9.57	-2.02	-10.13	

Notes: The dependent variable of the aggregate stock return (TTMK) at time t is R_t or sector i , $R_{i,t}$. The $\hat{\sigma}_{it}^2$ is conditional variance projected by a GED-GARCH(1,1) process; $\Delta Inf_t^+ = \Delta Inf_t > 0$ which is a rise in inflation; while $\Delta Inf_t^- = \Delta Inf_t < 0$, a decline in inflation; Δm_t is the money growth; $\Delta emv_{inf,t}$ is the US equity market volatility attributable to inflation risk; Δmpu_t is a change in monetary policy uncertainty; $dy_{i,t-1}$ is the lagged dividend yield in natural log in sector i ; D_{Crisis} is the dummy variable, which is set to unity for the 2008 world finance crisis and zero otherwise; $D_{COVID-19}$ is the dummy variable, which is set to unity for the COVID-19 period and zero otherwise. Numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

5.2.3. Time lag of inflation changes

As inflation rises, it is unclear whether market participants would react immediately by shorting their stocks, since rational traders are likely to wait longer in order to obtain more information and delay exercising any options until a more desirable time to sell off their stocks (Pindyck, 1991). This real options theory implies a delayed response to the inflation information. Estimates by adding the lagged terms of inflation factors, ΔInf_{t-1}^+ and ΔInf_{t-1}^- into the model are reported in Table 9.¹¹

The estimated results in Table 9 are consistent with our anticipation. More specifically, the aggregate data consistently show the delayed effect for ΔInf_{t-1}^+ and ΔInf_{t-1}^- . At the sectoral level, however, there are minor changes, notably that both ΔInf_t^+ and ΔInf_t^- show comparable qualitative results. However, in addition to the ENGy sector, estimates for the HLCA and TECH sectors now become positive in ΔInf_t^+ but are only significant at the 10% level. Yet, when bad news of inflation (ΔInf_t^+) hits market, the negative effects flow into next period for all sectors.

On the other hand, when good news of inflation (ΔInf_t^-) develops, the evidence consistently has a positive effect on the current stock returns (except for the CNSG sector). However, for the delayed effects, the coefficients of ΔInf_{t-1}^- , are either positive or display reverse signs, reflecting different responses for the lagged effect of lower inflation. With respect to the other independent variables, the estimated coefficients show very comparable and stable results.

6. Nonlinear specification

It is generally recognized that policymakers or market participants are continually assessing market conditions, and their decisions are sensitive to the emerging state variables (Lucas, 1976). This section tests the impact of the interaction of Δmpu_t with $\Delta emv_{inf,t-1}$. The following specification reflects this thought process.

$$R_{i,t} = C + \beta_1 \hat{\sigma}_{i,t}^2 + \beta_2 Inf_t + \beta_3 \Delta m_t + \beta_4 \Delta mpu_t + \beta_5 \Delta mpu_t \bullet \Delta emv_{inf,t-1} + \beta_6 dy_{i,t-1} + \beta_7 D_{Crisis} + \beta_8 D_{COVID-19} + \varepsilon_{i,t} \quad (7)$$

A special feature of equation (7) is that Δmpu_t from economic agents are assumed to react to lagged inflation-induced equity market volatility, $\Delta emv_{inf,t-1}$, (Lucas, 1976; Baker et al. 2022) which is reflected in the term of $\Delta mpu_t \bullet \Delta emv_{inf,t-1}$. The use of the lagged $\Delta emv_{inf,t-1}$ is based on the premise that the time lag for the stock market's reaction at the time of announcement of the inflation as well as investors' recognition lag.¹² Estimates of equation (7) are reported in Table 10.

Turning to the estimated results, whether those from aggregate or sectoral market, we find that the estimated parameters are stable and present a very comparable performance. For instance, the coefficients of $\{\hat{\sigma}_{i,t}^2, Inf_t, \Delta m_t, \Delta mpu_t, dy_{i,t-1}, D_{Crisis}$ and $D_{COVID-19}\}$ all have their anticipated signs and are statistically significant. The exceptions are ENGy, INVS and TECH sectors, where the coefficients for these stocks are significantly positive, supporting the notion that they can act as a hedge against inflation. A rather striking result of the estimation is the nonlinear term, where we find the coefficients of $\Delta mpu_t \bullet \Delta emv_{inf,t-1}$ are all negative. The effect on underlying behavior of these interactions is complex since the policymakers' decision-making process and investors' psychological reactions toward $\Delta emv_{inf,t-1}$ are subjective and unpredictable. However, this nonlinear term enhances the simple policy effect from Δmpu_t alone. When the total market (TTMK) is used, for example, the total effect of change in monetary policy uncertainty is $(\Delta mpu_t + \Delta mpu_t \bullet \Delta emv_{inf,t-1}) = -0.0159 - 0.0004$. It is the second (nonlinear) term that produces an augmented effect to the policy change. This augmented effect is also shown in the sectoral markets.

7. Conclusions and summary

This study extends the investigation of the asset-pricing model by incorporating inflation as a risk factor. Following the traditional CAPM model (as generated by a GARCH-type model) that posits stock returns are related to the conditional variance, this paper provides supporting evidence for the risk-return trade-off hypothesis consistent with the findings by French et al. (1987), Bali and Peng (2006), Bali and Engle (2010) and Chiang et al. (2015).

This paper finds significant evidence that money growth has a positive effect on stock market movements due to a liquidity effect and wealth effect (Tobin, 1969, 1982). This finding is consistent with the performance of stock price path as a V-shape (Chiang et al., 2007) for the GFC in the period between 2008 and 2009 as well as the COVID-19 crisis from beginning 2020 to 2021 (Chiang, 2022; 2023).

This study finds evidence to support the monetary policy uncertainty premium hypothesis, where heightened uncertainty causes a stock market selloff and leads to decline in returns. The result is consistent with the findings for US data (Husted et al., 2020) and US, EU, China and Japan data (Chiang, 2020) as well as data for the European countries (Batten et al., 2022).

This paper tests whether stocks can be used to hedge against inflation. Evidence from the aggregate market is not supportive of this contention; however, the sectoral sectors of BMAT, ENGy, TECH, and sometimes INVS, do have the capacity to hedge against inflation, since the estimated coefficients show positive signs for the coefficients of inflation. However, most sectoral stocks exhibit a negative

¹¹ The estimates of the coefficients on the variance equation are suppressed in order to fit in the table but will be available upon request. Likewise, lags can be added to the other variables. However, the aim of this paper is to highlight the inflation effect and adding more lags may lead to excess parametrization.

¹² It is approximately one month after the price data are collected by the Bureau of Labor Statistics as noted by Schwert (1981).

Table 10

Estimates of stock returns in relation to conditional variance, inflation, money growth, monetary policy uncertainty, and nonlinear term while controlling for dividend yield, dummies for finance crisis and COVID-19 shocks.

Sector	C	$\hat{\sigma}_{it}^2$	Inf_t	Δm_t	Δmpu_t	$\Delta mpu_t \bullet \Delta emv_{inf,t-1}$	dy_{it-1}	D_{Crisis}	$D_{COVID-19}$	$\omega_{i,0}$	ε_{it-1}^2	σ_{it-1}^2	$\bar{R}^2$
TTMK	5.7498	-17.0778	-2.5499	0.0539	-0.0159	-0.0004	1.6433	-8.3744	-5.4722	9.1315	0.1094	0.5821	0.04
	6.30	-4.92	-13.50	6.78	-24.53	-12.08	7.75	-7.98	-10.04	7.73	8.86	16.31	
FINL	2.8235	-9.4045	-1.6666	0.2621	-0.0201	-0.0004	1.5221	-9.8489	-15.4019	16.6352	0.1347	0.5848	0.06
	2.72	-2.19	-5.07	2.48	-12.40	-7.49	4.01	-7.13	-9.54	2.92	3.69	5.83	
HLCA	2.6672	-8.6764	-1.1095	0.3066	-0.0081	-0.0001	3.3685	-5.3574	-6.3510	9.4550	0.1270	0.5881	0.01
	2.12	-1.98	-8.77	6.02	-9.93	-3.23	11.02	-6.55	-4.27	3.36	2.45	6.19	
INDU	2.9463	-9.8751	-2.1122	0.7415	-0.0042	-0.0002	1.7980	-17.2184	-10.0204	13.6216	0.1391	0.5831	0.05
	3.65	-3.20	-6.95	5.81	-5.86	-6.46	6.08	-12.85	-16.79	3.56	4.09	6.61	
REIT	0.3373	-4.3798	-0.6452	0.4799	-0.0115	-0.0003	2.8442	-3.5049	-7.2143	13.1277	0.4491	0.5477	0.04
	4.83	-6.16	-2.11	5.21	-7.69	-4.23	6.52	-1.11	-2.16	1.40	1.68	2.40	
CNSG	2.2244	-9.6352	-1.3572	0.7830	-0.0166	-0.0002	3.3614	-8.7843	-9.4050	11.9489	0.1799	0.5628	0.11
	2.39	-2.70	-4.90	7.26	-9.01	-3.68	5.63	-7.52	-3.49	2.23	2.48	3.98	
RTAL	1.0079	-2.2453	-1.4716	0.6719	-0.0160	-0.0003	1.6018	-3.6454	-5.8424	16.0442	0.4489	0.5877	0.03
	1.87	-0.99	-3.81	4.99	-7.47	-3.83	5.11	-7.20	-10.50	1.34	3.12	3.17	
BMAT	1.6040	-9.9327	-1.1959	0.9977	-0.0027	-0.0004	5.2166	-12.6169	-9.7341	21.3798	0.2202	0.6130	0.08
	3.17	-4.79	-7.20	6.43	-6.19	-5.43	10.86	-15.63	-2.41	4.89	5.67	6.35	
ENGY	1.9750	-7.7010	0.6550	0.6082	-0.0196	-0.0002	0.9163	-6.4068	-16.9824	22.6363	0.1187	0.5959	0.07
	2.26	-2.04	12.06	9.87	-23.32	-10.00	5.81	-5.32	-2.30	3.03	3.88	5.89	
INVS	1.9953	-8.2881	0.6539	0.3129	-0.0330	-0.0004	2.6839	-5.2856	-14.2078	44.1030	0.1776	0.6284	0.04
	2.16	-1.72	5.97	2.19	-30.62	-8.48	11.65	-3.11	-18.45	1.81	2.98	4.12	
TECH	0.4343	-0.2983	1.5298	0.2564	-0.0245	-0.0003	0.9082	-4.9181	-7.5871	7.4453	0.4406	0.6783	0.04
	5.61	-0.92	3.88	1.32	-9.20	-3.28	3.76	-2.36	-4.05	1.32	1.97	5.02	

Notes: $R_{it} = C + \beta_1 \hat{\sigma}_{it}^2 + \beta_2 Inf_t + \beta_3 \Delta m_t + \beta_4 \Delta mpu_t + \beta_5 \Delta mpu_t \bullet \Delta emv_{inf,t-1} + \beta_6 dy_{it-1} + \beta_7 D_{Crisis} + \beta_8 D_{COVID-19} + \varepsilon_{it}$.

$R_{it}(R_{it})$ is the stock return for aggregate market (or sector i), $\hat{\sigma}_{it}^2$ is the conditional variance projected by a GED-GARCH(1,1) process; Inf_t is inflation; Δm_t is money growth; Δmpu_t is a change in monetary policy uncertainty; $\Delta emv_{inf,t}$ is the US equity market volatility attributable to inflation risk; dy_{it-1} is the lagged dividend yield in natural log in sector i ; D_{Crisis} is the dummy variable, which is set to unity for the 2008 world finance crisis and zero; otherwise; $D_{COVID-19}$ is the dummy variable, which is set to unity for the COVID-19 period and zero otherwise. Numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

Table A1

Regression estimates of G7 stock returns in relation to conditional variance and inflation while controlling risk and money growth.

Country	C	$\hat{\sigma}_{j,t}^2$	$Inf_{j,t}$	$\Delta emv_{inf,t}$	$\Delta m_{j,t}$	Δmpu_t	$\Delta dy_{j,t-1}$	D_{Crisis}	$D_{COVID-19}$	$\omega_{j,0}$	$\varepsilon_{j,t-1}^2$	σ_{it-1}^2	$\bar{R}^2$
US	-8.7734	2.5478	-0.0691	-0.0238	0.3442	-0.0126	2.2465	-13.4914	-9.0643	9.9174	0.2191	0.5973	0.08
	-8.66	11.50	-3.70	-10.61	3.10	-6.39	9.39	-17.82	-4.14	6.34	5.08	11.19	
CA	-18.6632	6.4573	-0.4397	-0.0208	-1.2436	-0.0123	0.5913	-8.6002	-6.5821	8.4992	0.1047	0.5647	0.09
	-7.84	9.22	-3.13	-13.01	-7.64	-6.24	5.50	-8.73	-7.30	5.29	3.47	9.32	
FR	-2.4022	0.7541	-0.3033	-0.0282	0.1252	-0.0239	0.8618	-5.9358	-8.2909	14.9433	0.1667	0.6112	0.10
	-0.64	0.78	-2.18	-17.46	3.27	-10.34	2.77	-4.99	-2.68	1.85	0.88	3.74	
GM	-4.9729	1.4552	-0.6308	-0.0276	-0.2203	-0.0235	1.3502	-5.7124	-9.8016	16.4392	0.1643	0.6004	0.05
	-1.68	2.03	-2.67	-11.30	-2.28	-8.12	4.18	-3.91	-3.48	3.09	4.87	7.98	
IT	-9.0065	1.9046	-1.3475	-0.0191	-0.0585	-0.0220	2.5848	-8.9787	-5.7096	4.5529	0.0914	0.8172	0.07
	-1.65	1.32	-1.97	-3.18	-0.54	-3.64	3.53	-5.30	-1.88	1.33	1.33	8.18	
UK	-5.3170	1.7848	-0.3372	-0.0251	0.3905	-0.0105	1.6577	-8.6802	-10.9224	8.7310	0.1629	0.5909	0.07
	-5.68	6.75	-3.17	-10.64	3.19	-5.28	6.29	-8.19	-4.70	3.84	5.83	6.09	
JP	-8.4828	2.4770	-0.4099	-0.0271	0.6153	-0.0250	1.0811	-8.8581	-10.2870	14.8538	0.1224	0.5733	0.09
	-1.83	2.08	-7.93	-9.85	9.06	-9.69	8.21	-8.22	-3.62	2.15	4.65	4.10	

Notes: The dependent variable is stock return for country j at time t , $R_{j,t}$. Numbers in the first row are the estimated coefficients, the second row contains the z-statistics. The critical values of t-distribution at the 1%, 5%, and 10% levels of significance are 2.63, 1.98, and 1.66, respectively. $\bar{R}^2$ is the adjusted R-squared.

relation for stock returns and inflation, supporting the findings in literature documented by Fama (1981), Geske and Roll (1983). By alternating the inflation measures with expected and unexpected inflation rates, this study yields similar results, although the signs for the coefficients for the INDU and REIT sectors turn to positive using an unexpected inflation rate. This result can be caused by spurious correlations.

Dividing the inflation series into positive change vs. negative change regimes produces evidence that confirms that a rise in inflation causes stock returns to decline. The only exception is stocks in the ENGY sector, which shows a positive sign and is highly significant, suggesting that only ENGY stocks can be used to hedge against inflation risk. This is consistent with the evidence by Bouri et al. (2023). However, as good news of declining inflation develops, all sectoral stock returns shift upward. Thus, this two-regime model provides solid evidence supporting the inverse stock returns-inflation relation, the only exception is the ENGY sector.

This study also identifies a new factor, equity market volatility, which positively covaries with inflation. The evidence shows that a rise in $\Delta mv_{inf,t}$ triggered by higher inflation would produce fears that cause a selloff and, hence, lower stock returns. Once again, the exception is the ENGY sector, which has a positive effect (Table 8). The significance of an inflation-volatility factor as explored in this study has not been incorporated in models in previous studies.

In summary, this paper successfully integrates risk/uncertainty factors, $\{\hat{\sigma}_{it}^2, \Delta mpu_t\}$, economic fundamental $\{dy_{t-1}\}$, inflation elements $\{\Delta Inf_t^+, \Delta Inf_t^-, \Delta mv_{inf,t}, \Delta m_t\}$, interacting term $\{\Delta mpu_t \bullet \Delta mv_{inf,t-1}\}$ and dummies from external shocks into a unified model, which sheds significant light on the relation between stock returns and inflation in the short run.

This study has important policy implications. Evidence indicates that monetary policy interacts with inflation and hence produces inflation-induced market volatility. The findings are consistent with the Lucas critique (1976). It proposes that monetary authority or an economic agent tends to dynamically react to state variables as shown in the interacting term, which enhances the effect of a change in monetary policy uncertainty on stock returns. Further, evidence strongly suggests that growth in the money supply will have a positive effect on stock returns due to an injection of liquidity. However, this study also concludes that a rise in monetary policy uncertainty could increase investors' fears, which could have an adverse effect on stock returns. One crucial implication is the importance of stable monetary growth as a precondition for maintaining market stability as advocated by Milton Friedman (1948; 1968) and recent observed by Beckworth (2022).¹³

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix. Regression estimates of stock returns in response to inflation for GED-GARCH estimates of G7

This Appendix reports the estimates for G7 countries by extending the data to April 2023 and is compatible to model 6 in Table 4. The model describes that aggregate stock return for country j , $R_{j,t}$, is assumed to be explained by its own factors, $\hat{\sigma}_{j,t}^2$, $Inf_{j,t}$, $\Delta m_{j,t}$ and $dy_{j,t-1}$ as well as US factors, $\Delta mv_{inf,t}$ and Δmpu_t . The latter two variables are related to uncertainty from US spillovers. Estimates of the results are comparable and statistically significant; these results apply to the estimates of the coefficients for $\hat{\sigma}_{j,t}^2$, $Inf_{j,t}$, $dy_{j,t-1}$, $\Delta mv_{inf,t}$ and Δmpu_t . Only the coefficients of $\Delta m_{j,t}$ for CA, GM and IT have the opposite signs. In general, the evidence supports the finding that inflation has a strong negative effect on stock returns; uncertainty from the US market will also produce an adverse effect. The performance of G7 countries are consistent with the study by Chiang (2019), who, however, did not include inflation and growth of money supply in the test equation.

Table A1

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¹³ Milton Friedman noted financial stability issues primarily through the lens of nominal income and the money stock (Friedman, 1968). His notion suggests that if both were kept on a stable growth path, then financial stability would generally follow (Beckworth, 2022). While monetary authority manipulates policy to deal with the unexpected shocks, the Friedman's rule should be viewed as a guiding principle.

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